

ZeroTick Benchmark's

Dell Vostro 3401

Specifications: Intel i3 1000g1
16 GB

Phase 1: Pure Math & CPU Throughput

```
=== RUNNING LOCAL QUANT & MATH ENGINE BENCHMARK ===  
Processed 10000000 sequential ticks in 73.117573ms  
Math Throughput : 136.77 million ticks/sec  
Mean Price : 5099.9995  
Std Deviation : 2886.751490  
Upper Bollinger : 10873.5025  
Lower Bollinger : -673.5035  
Max Drawdown : 0.0000%  
Memory Overhead : O(1) stack space (72 bytes)
```

Phase 2: Deep OS Profiling

Flame Test:

```
=== RUNNING LOCAL QUANT & MATH ENGINE BENCHMARK ===  
Processed 10000000 sequential ticks in 77.915888ms  
Math Throughput : 128.34 million ticks/sec  
Mean Price : 5099.9995  
Std Deviation : 2886.751490  
Upper Bollinger : 10873.5025  
Lower Bollinger : -673.5035  
Max Drawdown : 0.0000%  
Memory Overhead : O(1) stack space (72 bytes)  
[ perf record: Woken up 1 times to write data ]  
[ perf record: Captured and wrote 0.020 MB perf.data (308 samples) ]
```

Memory Footprint:

```
=== RUNNING LOCAL QUANT & MATH ENGINE BENCHMARK ===
Processed 10000000 sequential ticks in 68.756133ms
Math Throughput : 145.44 million ticks/sec
Mean Price : 5099.9995
Std Deviation : 2886.751490
Upper Bollinger : 10873.5025
Lower Bollinger : -673.5035
Max Drawdown : 0.0000%
Memory Overhead : 0(1) stack space (72 bytes)
  Command being timed: "target/release/zerotick bench"
  User time (seconds): 0.06
  System time (seconds): 0.00
  Percent of CPU this job got: 97%
  Elapsed (wall clock) time (h:mm:ss or m:ss): 0:00.06
  Average shared text size (kbytes): 0
  Average unshared data size (kbytes): 0
  Average stack size (kbytes): 0
  Average total size (kbytes): 0
  Maximum resident set size (kbytes): 2280
  Average resident set size (kbytes): 0
  Major (requiring I/O) page faults: 0
  Minor (reclaiming a frame) page faults: 93
  Voluntary context switches: 0
  Involuntary context switches: 0
  Swaps: 0
  File system inputs: 0
  File system outputs: 0
  Socket messages sent: 0
  Socket messages received: 0
  Signals delivered: 0
  Page size (bytes): 4096
  Exit status: 0
```

Throughput Consistency:

Benchmark 1: target/release/zerotick bench

```
Time (mean ± σ):      71.0 ms ± 1.6 ms   [User: 69.0 ms, System: 1.8 ms]
Range (min ... max):  69.8 ms ... 77.7 ms 39 runs
```

MacBook Pro

Specifications: M5
16 GB

Phase 1: Pure Math & CPU Throughput

```
Finished `release` profile [optimized] target(s) in 0.00s
Running `target/release/zerotick bench`
=== RUNNING LOCAL QUANT & MATH ENGINE BENCHMARK ===
Processed 10000000 chaotic ticks in 69.974083ms
Math Throughput : 142.91 million ticks/sec
Mean Price : 99.9500
Std Deviation : 0.018202
Upper Bollinger : 99.9864
Lower Bollinger : 99.9136
Max Drawdown : 0.1488%
Memory Overhead : 0(1) stack space (72 bytes)
```

Phase 2: Deep OS Profiling

Flame Test:

```
=== RUNNING LOCAL QUANT & MATH ENGINE BENCHMARK ===
Processed 10000000 chaotic ticks in 69.436583ms
Math Throughput : 144.02 million ticks/sec
Mean Price : 99.9500
Std Deviation : 0.018202
Upper Bollinger : 99.9864
Lower Bollinger : 99.9136
Max Drawdown : 0.1488%
Memory Overhead : 0(1) stack space (72 bytes)
All tasks terminated.
Local server listening at http://127.0.0.1:3000
Press Ctrl+C to stop.
```

Memory Footprint:

```
=== RUNNING LOCAL QUANT & MATH ENGINE BENCHMARK ===
Processed 10000000 chaotic ticks in 69.311458ms
Math Throughput : 144.28 million ticks/sec
Mean Price : 99.9500
Std Deviation : 0.018202
Upper Bollinger : 99.9864
Lower Bollinger : 99.9136
Max Drawdown : 0.1488%
Memory Overhead : 0(1) stack space (72 bytes)
      0.07 real      0.07 user      0.00 sys
      1769472 maximum resident set size
           0 average shared memory size
           0 average unshared data size
           0 average unshared stack size
       279 page reclaims
           0 page faults
           0 swaps
           0 block input operations
           0 block output operations
           0 messages sent
           0 messages received
           0 signals received
           0 voluntary context switches
           4 involuntary context switches
    486456534 instructions retired
    161856644 cycles elapsed
     1048912 peak memory footprint
```

Throughput Consistency:

```
Benchmark 1: target/release/zerotick bench
Time (mean ± σ): 69.5 ms ± 0.2 ms [User: 68.1 ms, System: 1.1 ms]
Range (min ... max): 69.1 ms ... 70.0 ms 40 runs
```